SwiftCart Retail E-Commerce Sales Performance

Introduction

SwiftCart is an e-commerce retailer selling Books, Clothing, Electronics, and Home products. This short case study reviews sales value, product pricing, order quantity, and customer ratings to identify practical business priorities.

Problem statement

SwiftCart needs to know which product categories are driving sales, whether customers are satisfied, and whether higher prices may be creating customer-value concerns.

Objectives

1. Measure overall sales performance.
2. Compare revenue and quantity across product categories.
3. Review the price distribution.
4. Compare customer ratings by category.
5. Check the relationship between price and customer rating.
6. Identify risks in the most expensive orders.

Dataset description

The dataset contains **20 orders** (Order IDs 1001–1020) across four product categories. It includes **Price**, **Quantity**, and **Customer_Rating**. Sales revenue was calculated as **Price × Quantity**.

Field	Description
Order_ID	Unique order number
Product_Category	Books, Clothing, Electronics, or Home
Price	Product price per order

Quantity	Units purchased
Customer_Rating	Rating from 1 to 5
Revenue	Calculated as Price × Quantity

Tools used

- **Python (Pandas):** data calculation and aggregation
- **Data Formulator:** charts and visual analysis

Tasks and results

Task 1 — Calculate overall sales performance

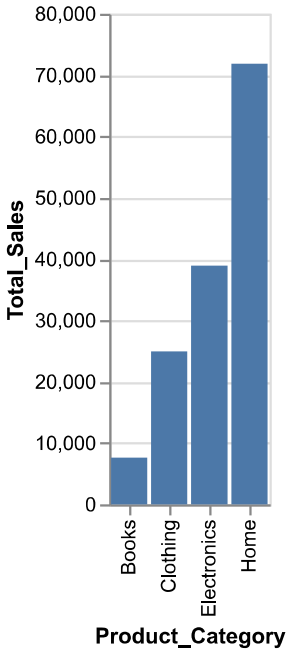
SwiftCart recorded **$143,323** in total revenue from **20 orders** and **61 units**. The average product price was **$2,334.05**, and the average customer rating was **3.86/5**.

Metric	Result
Total orders	20
Total units	61
Total revenue	$143,323
Average price	$2,334.05
Average customer rating	3.86 / 5

Task 2 — Identify the highest-revenue product category

Home generated the most revenue: **$71,874** (50.1% of total revenue), with **31 units** across 9 orders. Electronics ranked second at **$38,910**.

Category	Orders	Units	Revenue
Home	9	31	$71,874
Electronics	6	16	$38,910
Clothing	4	10	$24,963
Books	1	4	$7,576

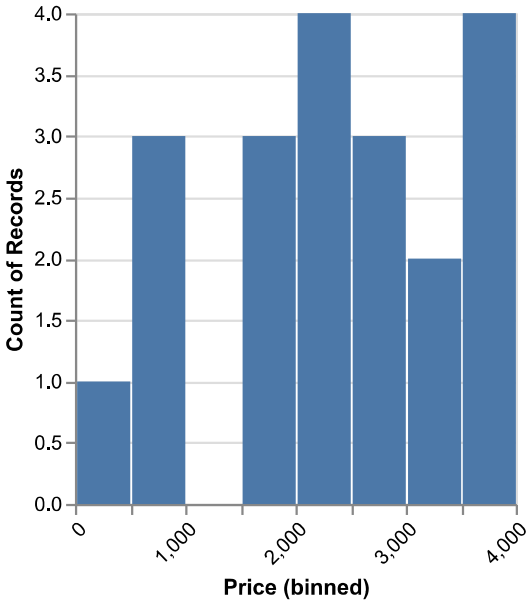


Task 3 — Examine product-price distribution

Prices are concentrated in the middle-to-high range: **13 of 20 orders** are priced from **$2,000 to $3,999**. Only four orders are below $1,000.

Price range	Orders
Below $1,000	4
$1,000–$1,999	3

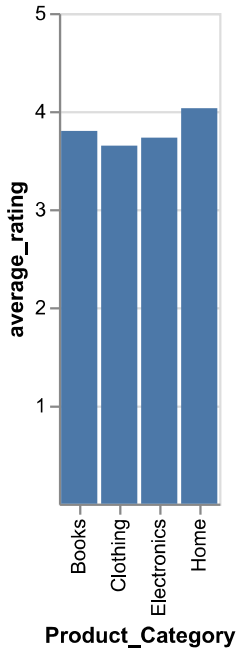
$2,000–$2,999	7
$3,000–$3,999	6



Task 4 — Compare customer satisfaction by category

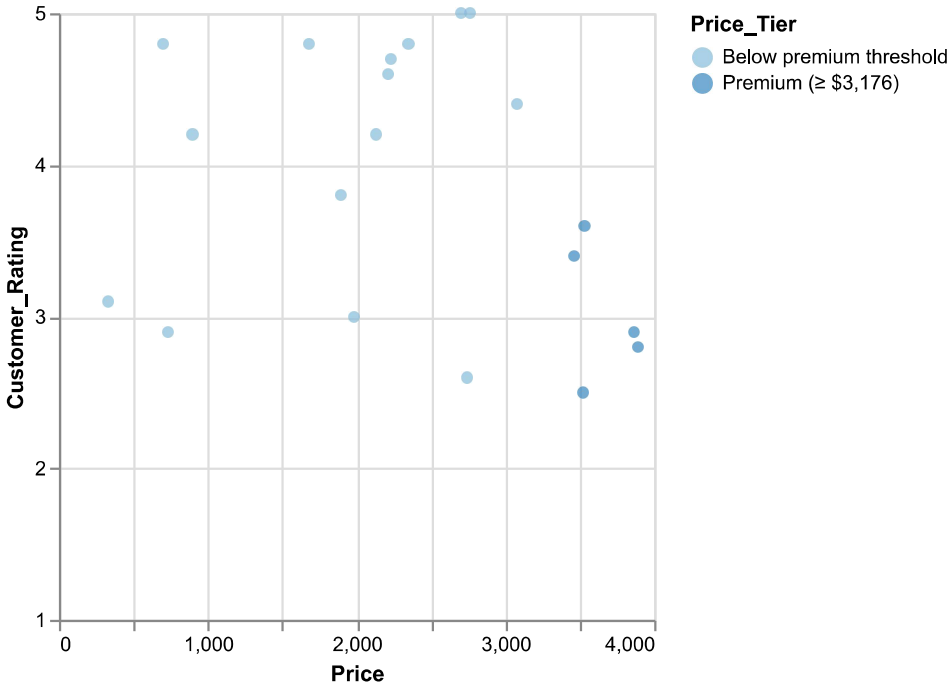
Home received the strongest average rating (**4.03/5**). Clothing had the lowest average rating (**3.65/5**), although it included one 5.0-rated order.

Category	Average rating
Home	4.03
Books	3.80
Electronics	3.73
Clothing	3.65



Task 5 — Review price versus customer rating

Higher prices did **not** lead to higher ratings in this small sample. The relationship between price and rating was weakly negative (**correlation: -0.23**), so price alone is not a reliable sign of customer satisfaction.



Task 6 — Assess premium-price order risk

The top 25% of prices begin at **$3,176**. These **5 premium-priced orders** generated **$51,344** in revenue, but their average rating was only **3.04/5**—well below the overall average of 3.86/5. This suggests a possible value, quality, product-description, or service issue for expensive items.

Premium-order metric	Result
Price threshold	$3,176 and above
Premium orders	5
Premium-order revenue	$51,344
Premium-order average rating	3.04 / 5

Key findings

- **Home is the best-performing category** for both revenue and satisfaction.
- **Electronics and Clothing need closer review** because their ratings are below the overall average.
- SwiftCart's assortment is mainly priced from **$2,000 to $3,999**.
- Premium-priced orders generate meaningful revenue, but they have **lower customer satisfaction**.
- Price should not be used alone to judge product performance; revenue and rating should be considered together.

Conclusion

SwiftCart should protect stock availability for Home products, especially its successful mid-priced items. It should investigate expensive Electronics and Home orders before expanding premium inventory, because customer ratings are weaker at the top price range. Clothing should be replenished carefully until its satisfaction level improves.

Limitation: This analysis is based on only 20 orders. Results are useful for direction-setting, but more sales data is needed before making major forecasting or inventory decisions.